IITG Racing

Virtual EV Challenge — Team Report

Conducted by

IPG Automotive and Bosch

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1. FMU Details

The Functional Mock-up Unit (FMU) was generated using MATLAB Simulink's FMI Export Toolbox, compliant with FMI 2.0 for Co-Simulation.

1.1 Parameters Used in the FMU

Parameter	Symbol	Value	Units
Air Density	ϱ (rho)	1.225	kg/m^3
Drivetrain Efficiency	η (eta)	0.90	—
Gravitational Acceleration	g	9.81	m/s^2
Frontal Area	A	2.48	m^2
Vehicle Mass	m	1,400	kg
Rolling Resistance Coefficient	C_{rr}	0.01	—
Drag Coefficient	C_n	0.31	—
Maximum Vehicle Speed	V_{max}	16.667	m/s
Gear Ratio	G	6	—

2. Phase 1 — Optimal Speed and Motor Force Modelling

2.1 Optimal Speed Model

The Simulink model computes the optimal vehicle speed (V_opt) to maximise energy efficiency over the route. The governing equation is:

$$V\_opt = \sqrt{\{ [(E/d)\cdot\eta - C_{rr}\cdot m\cdot g] / (0.5\cdot\varrho\cdot C_n\cdot A) \}} \quad \dots (1)$$

V_opt is constrained not to exceed the maximum vehicle speed of 16.667 m/s.

2.2 Motor Force Model

The motor force required to maintain the target speed is calculated as:

$$F_{motor} = C_{rr}\cdot m\cdot g + \tfrac{1}{2}\cdot\varrho\cdot C_n\cdot A\cdot v^2_{act} + m\cdot a \quad \dots (2)$$

2.3 Simulink Model — Block Diagram

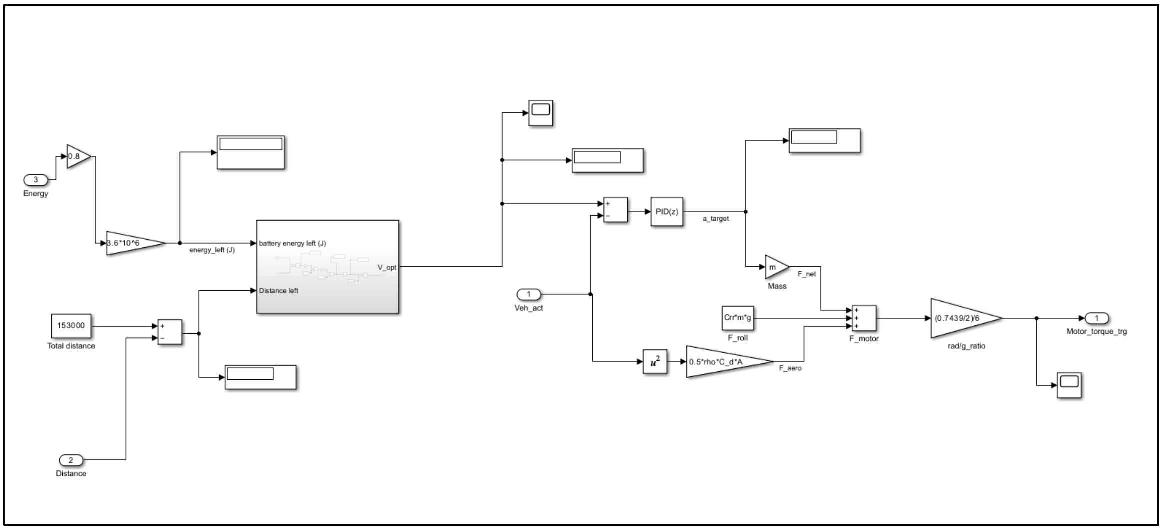


Figure 1: MATLAB Simulink FMU block diagram

2.4 Optimum Velocity Subsystem

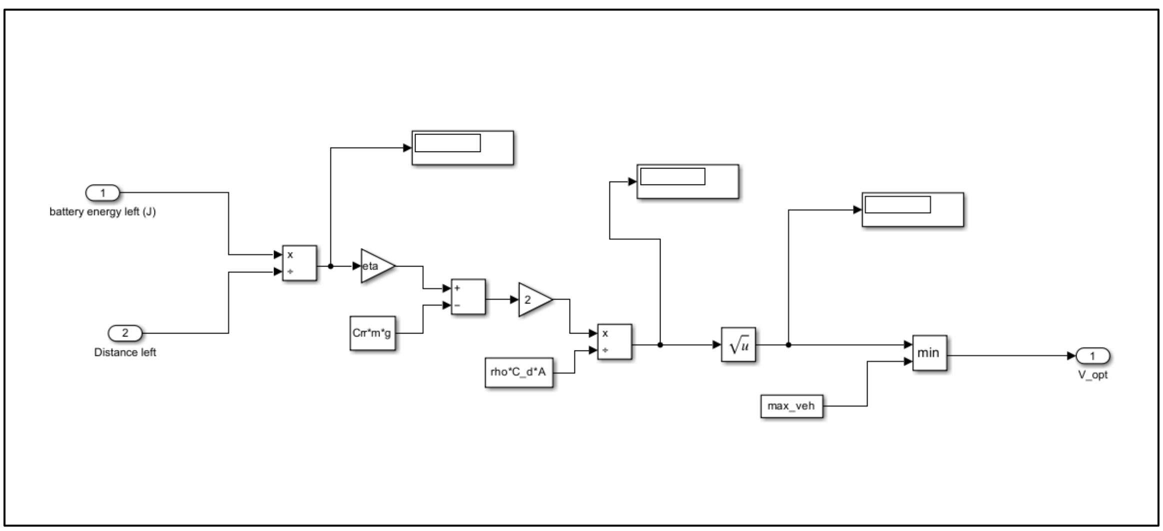


Figure 2: Optimum velocity subsystem in Simulink

2.5 Simulation Summary

Total distance covered: 153 km **Elapsed time:** 9,612.2 s (2.67 hours) **Residual SoC:** 19%

2.6 Velocity vs. Time

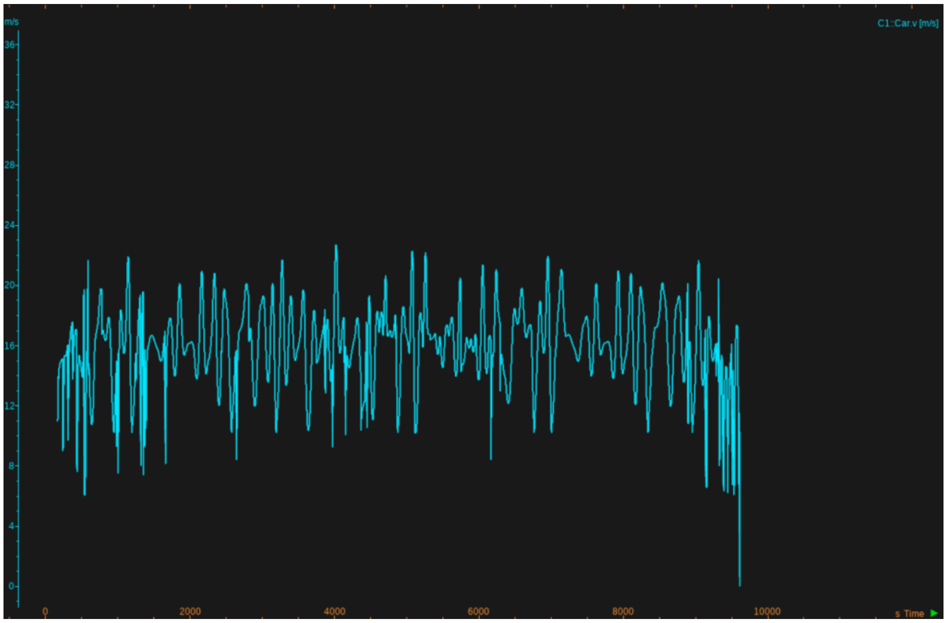


Figure 3: Vehicle velocity profile over the complete route

3. Phase 2 — AI-Integrated Charging Optimisation

3.1 System Architecture

3.1.1 Overview

The integrated system couples three primary components:

- CarMaker — high-fidelity vehicle dynamics and powertrain simulation engine (source of motor torque, total force, vehicle speed, distance travelled, and battery SoC).
- KUKSA Databroker (VSS) — middleware for publishing and subscribing to standardised Vehicle Signal Specification (VSS) signals.
- AI Agent System — a multi-agent optimiser (Requester and Optimizer agents implemented with uAgents) that consumes vehicle state from KUKSA, computes optimal charging decisions, and writes results back to KUKSA for CarMaker consumption.

Data flow (logical):

CarMaker → Bridge (FMU / PyCarMaker script) → KUKSA → AI Agents → KUKSA → Bridge → CarMaker

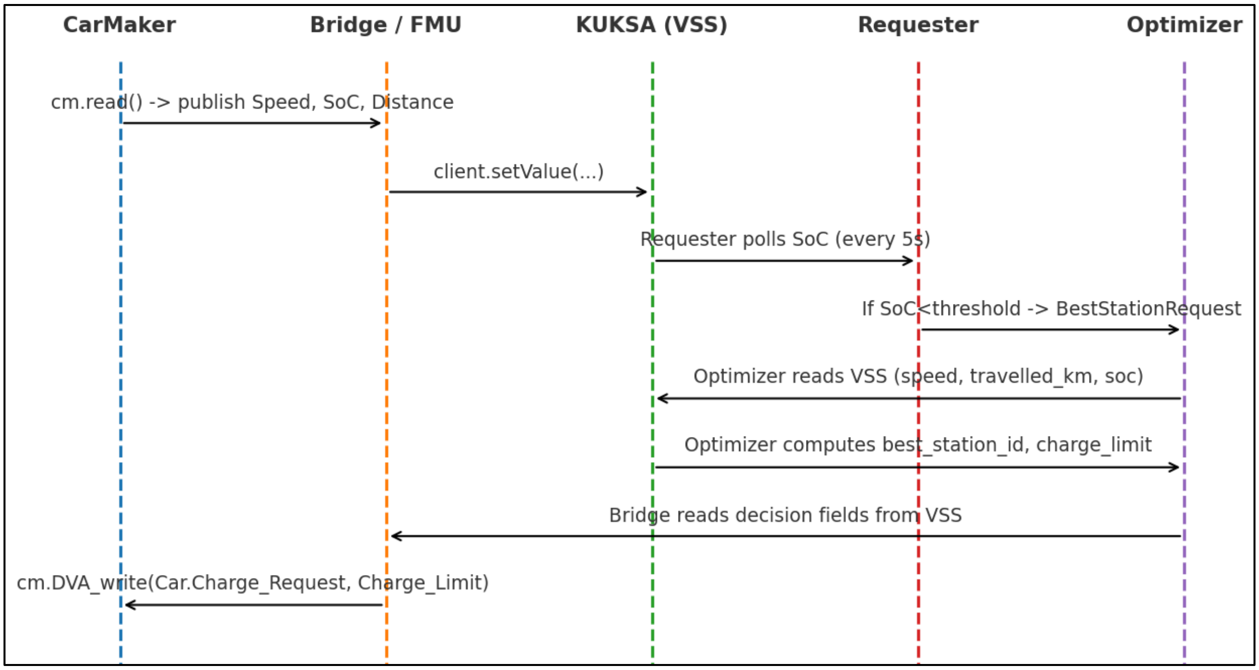


Figure 4: Additional velocity characteristics plot

3.1.2 Implementation Mapping

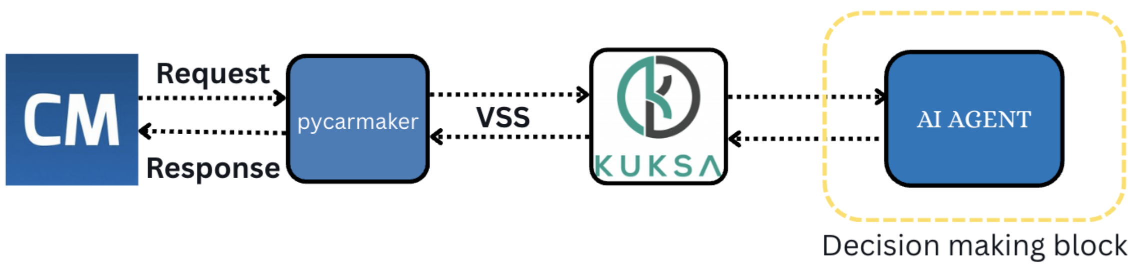


Figure 5: System data-flow diagram

The three integration pathways are implemented as follows:

- CarMaker → KUKSA (Bridge): Implemented using pycarmaker and kuksa_client. The bridge reads PT.BCU.BattHV.SOC, Car.Distance, and Car.v from CarMaker and pushes them into KUKSA via client.setValue().
- KUKSA → AI Agents: AI_Agents.py subscribes to VSS signals (Vehicle.Speed, Vehicle.TraveledDistance, Vehicle.Acceleration.Longitudinal) via the VSSSubscription helper.
- Decision Publishing: Car.Charge_Request activates the Optimizer Agent, which writes the selected best_station_id and Car.Charge_Limit back to KUKSA.
- Agent Orchestration: Bureau() runs both Optimizer and Requester agents concurrently.

3.2 AI Logic Design

3.2.1 Objective

The AI optimiser targets:

- Minimise total trip time (travel time + charging time).
- Guarantee safe reachability — the vehicle must always be able to reach the chosen charging station with SoC above MIN_SOC_ARRIVAL.
- Maintain a minimum residual SoC ≥ 20% at the end of the route.

The optimiser returns: (1) the best charging station ID, and (2) the required Charge Limit (target SoC).

3.2.2 Agent Interaction Design

The AI agents interact according to the following logic:

- AI Agent subscribes to live VSS values from KUKSA.
- **Requester Agent monitors SoC continuously.**
    - If current SoC < 25%, the Requester Agent triggers the Optimizer Agent via a message.
- Optimizer Agent evaluates candidate stations on four criteria:
    - Travel time to each station.
    - Predicted SoC on arrival (must be ≥ 15%).
    - Travel time from the station to the destination.
    - Predicted SoC at the destination.
- Charge Limit is calculated to minimise charging time while guaranteeing a safe final SoC.
- Selected station_id and Charge_Limit are published to KUKSA via VSS for CarMaker consumption.

3.3 Bridge Highlights

3.3.1 Purpose and Role

The PyCarMaker bridge provides real-time data translation between CarMaker and KUKSA. Its primary responsibilities are:

- Temporal alignment of simulation state and AI decisions.
- Bidirectional signal translation between CarMaker quantities and VSS paths.
- Reliable event-driven triggering of charging decisions based on live SoC readings.

3.4 Performance Metrics and Evaluation Plan

The table below lists recommended performance metrics, their measurement methods, and target values for simulation-grade evaluation.

Metric	Definition & Measurement	Target
End-to-End Latency	Time from CarMaker state publish to optimizer decision write-back to CarMaker	< 200 ms
Decision Correctness	Fraction of feasible optimizer decisions; SoC at arrival ≥ MIN_SOC_ARRIVAL; final SoC ≥ MIN_SOC_FINISH	100%
SoC Estimation Error	Mean Absolute Error between AI-estimated and CarMaker ground-truth SoC	MAE < 2–5 pp
Optimization Overhead	Wall-clock time for optimizer to evaluate all candidate stations per request	< 100 ms

Robustness / Stability	Zero crashes, no missed messages, no deadlocks across extended drive-cycle scenarios	Zero errors
Scalability	Latency and CPU use vs. number of stations/vehicles	Linear or sub-linear growth

4. Team Experience and Lessons Learned

4.1 Phase 1 — Simulink Modelling

- Built the vehicle and powertrain model in MATLAB Simulink.
- Optimised motor torque for energy efficiency while meeting performance targets.
- Validated the energy consumption baseline for charging calculations.

4.2 Phase 2 — AI Integration

- Connected CarMaker/Simulink signals (SoC, speed, distance) to KUKSA VSS.
- Designed AI agents (Requester + Optimizer) for charging-station selection.
- Returned optimal charging station ID and charge target to the simulator in real time.

4.3 Lessons Learned

- Torque-based SoC estimation is more reliable than distance-based methods.
- Event-driven FMUs improve system responsiveness compared to polling loops.
- Clear and consistent signal mapping in VSS is critical for effective debugging.
- Separating monitoring (Requester) and computation (Optimizer) agents improves overall responsiveness.

5. Simulation Results

The following plots show the AI-driven charging optimisation results for three different initial State of Charge (SoC) scenarios.

5.1 Results — Initial SoC: 50%

```
22.22449493408203
INFO:     [requester]: SoC low (22.22%), requesting optimizer
INFO:     [optimizer]: Received best station request
INFO:     [requester]: Booking successful → Best station 14 with charge limit 41.70875
INFO:     [requester]: station_id=14 distance_to_station_km=18.82 time_to_station_h=0.29 soc_at_arrival_percent=15.17 charging_time_h=0.53 time_to_des
tination_h=0.89 final_soc_percent=20.0 total_trip_time_h=1.71
```

Figure 6a: Signal overview — 50% initial SoC

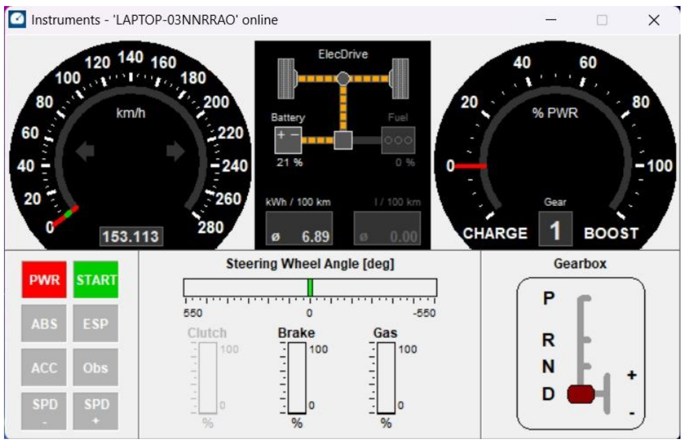


Figure 6b: Final SoC

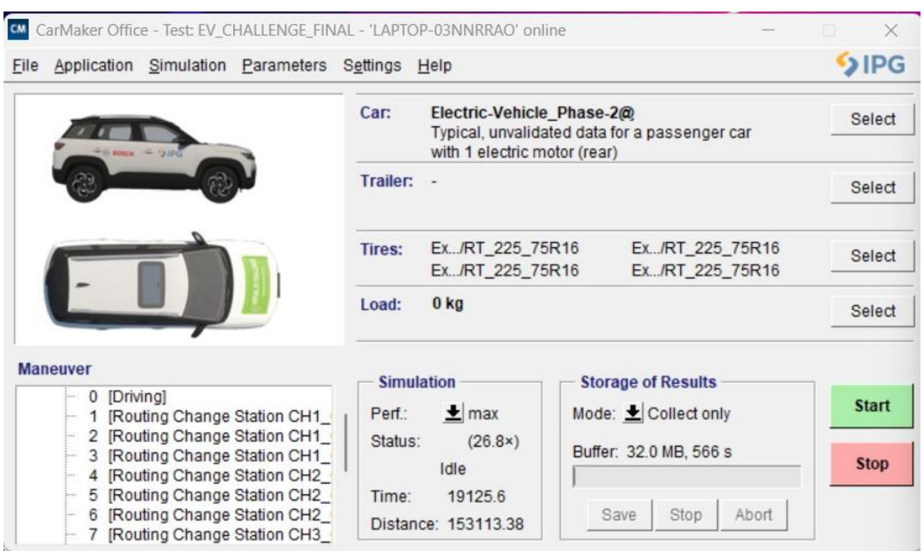


Figure 6c: Time and distance — 50% initial SoC

5.2 Results — Initial SoC: 40%

```
25.285921096801758
23.55013656616211
INFO:     [requester]: SoC low (23.55%), requesting optimizer
INFO:     [optimizer]: Received best station request
INFO:     [requester]: Booking successful → Best station 11 with charge limit 66.61125
INFO:     [requester]: station_id=11 distance_to_station_km=13.6 time_to_station_h=0.21 soc_at_arrival_percent=18.45 charging_time_h=0.96 time_to_dest
ination_h=1.5 final_soc_percent=30.0 total_trip_time_h=2.67
```

Figure 7a: Signal overview — 40% initial SoC

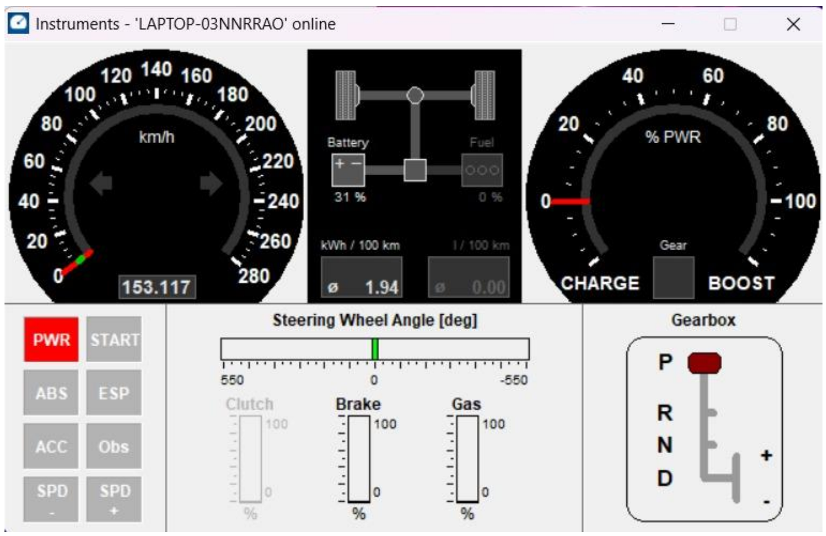


Figure 7b: Final SoC

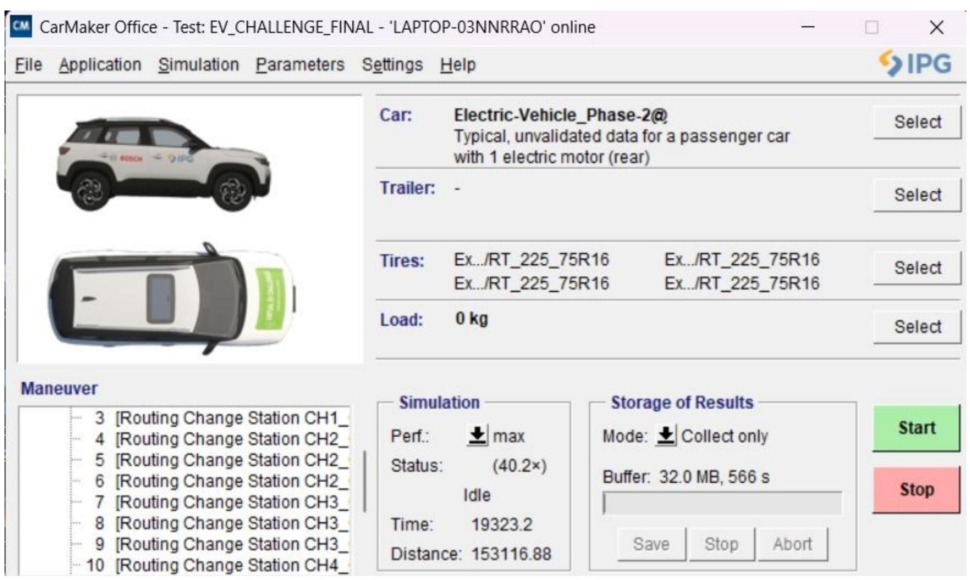
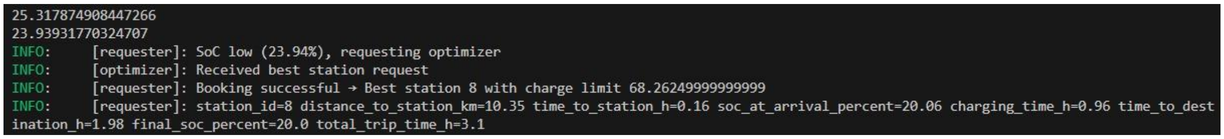


Figure 7c: Time and distance — 40% initial SoC

5.3 Results — Initial SoC: 30%



```
25.317874908447266
23.93931770324707
INFO:     [requester]: SoC low (23.94%), requesting optimizer
INFO:     [optimizer]: Received best station request
INFO:     [requester]: Booking successful → Best station 8 with charge limit 68.26249999999999
INFO:     [requester]: station_id=8 distance_to_station_km=10.35 time_to_station_h=0.16 soc_at_arrival_percent=20.06 charging_time_h=0.96 time_to_dest
ination_h=1.98 final_soc_percent=20.0 total_trip_time_h=3.1
```

Figure 8a: Signal overview — 30% initial SoC

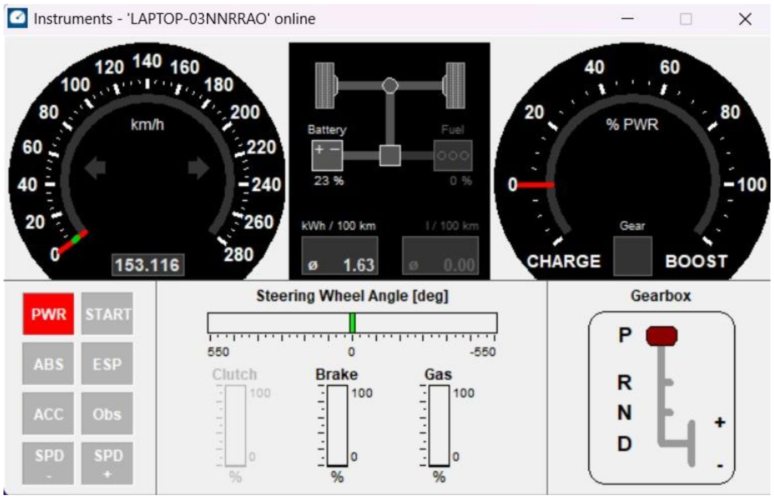


Figure 8b: Final SoC

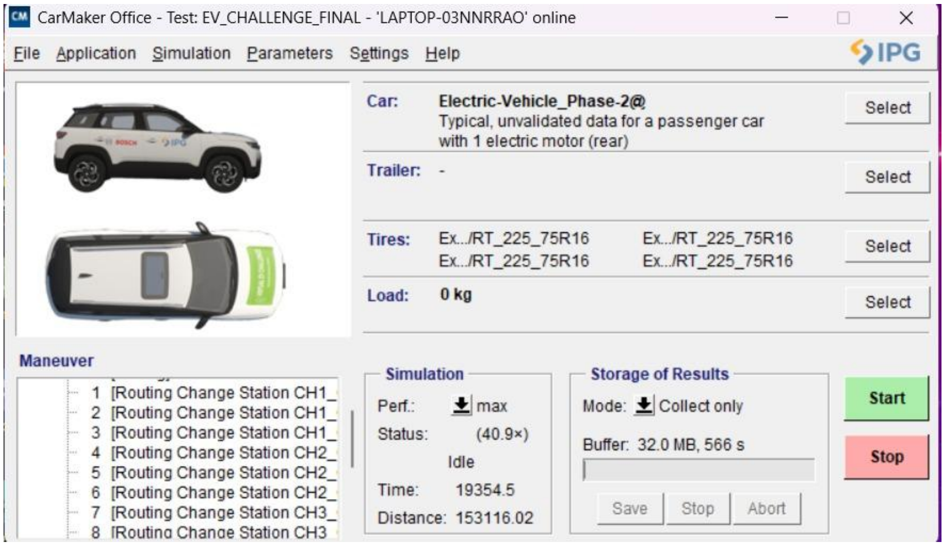


Figure 8c: Time and distance — 30% initial SoC

End of Report

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